

11. Oxalic acid is the common name for ethanedioic acid. It is found in a wide range of plants and when purified it can form a dihydrate,  $(\text{COOH})_2 \cdot 2\text{H}_2\text{O}$ . The concentration of a saturated solution of oxalic acid can be found by titration.

(a) A sample of  $25.0\text{ cm}^3$  of a saturated solution of oxalic acid is diluted to form  $250.0\text{ cm}^3$  of an aqueous solution. Samples of  $25.0\text{ cm}^3$  of this diluted solution were heated and titrated using acidified potassium manganate(VII) to oxidise the oxalic acid.

(i) The half-equation for the oxidation of oxalic acid is shown below.



State the half-equation for the reduction of manganate(VII) ions in acid solution and use it to write the equation for the oxidation of oxalic acid by acidified manganate(VII) ions. [2]

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- (ii) The titration of  $25.0\text{ cm}^3$  samples of the diluted oxalic acid solution against a potassium manganate(VII) solution of concentration  $0.0505\text{ mol dm}^{-3}$  gave the following results.

|                                                    | 1     | 2     | 3     | 4     |
|----------------------------------------------------|-------|-------|-------|-------|
| Volume of $\text{KMnO}_4$ solution / $\text{cm}^3$ | 31.65 | 31.30 | 31.60 | 31.75 |

- I. Use appropriate data to find the mean volume of  $\text{KMnO}_4$  solution used. [1]

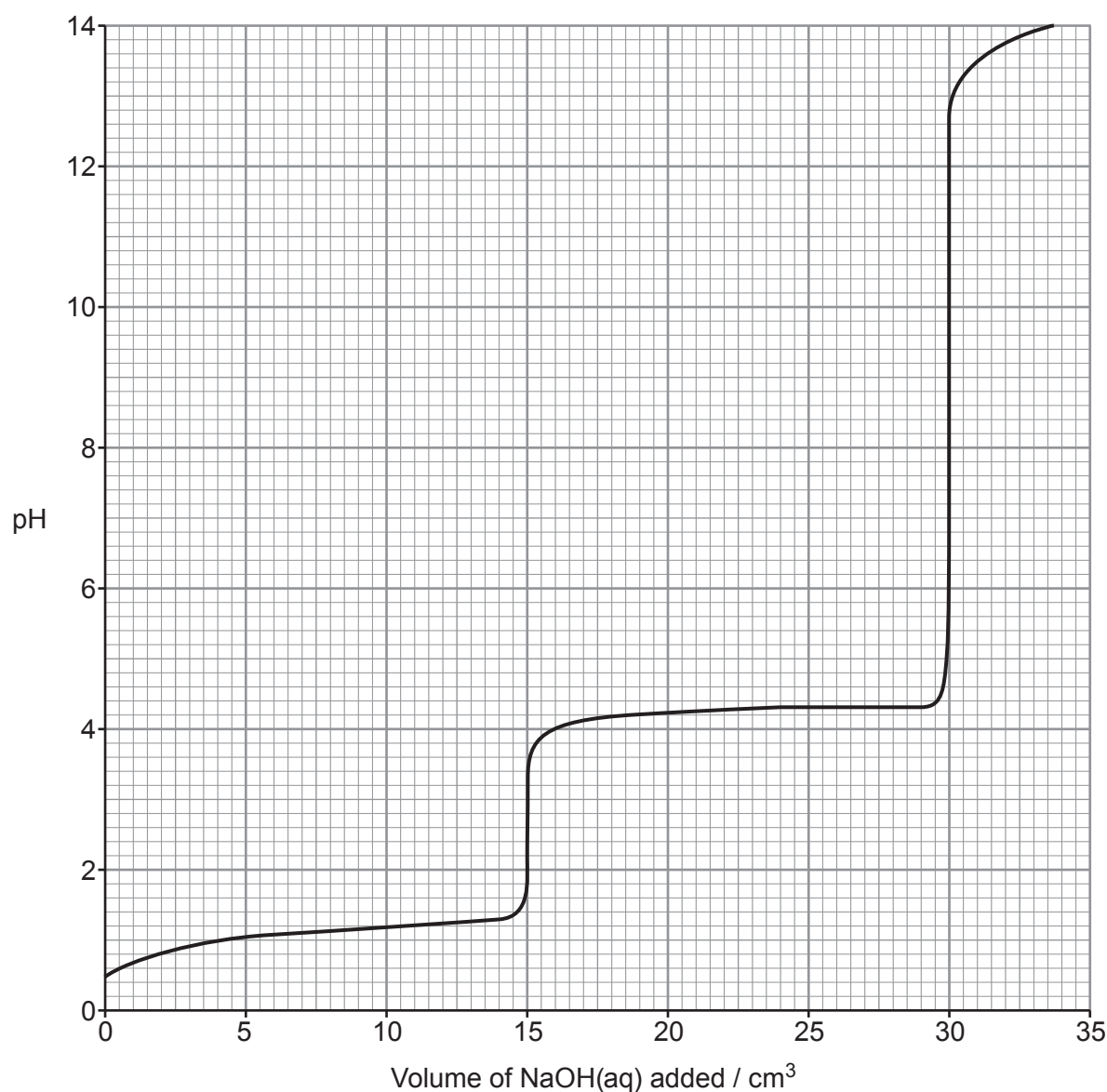
Mean volume = .....  $\text{cm}^3$

- II. Calculate the concentration of the original saturated solution of oxalic acid. [3]

Concentration = .....  $\text{mol dm}^{-3}$



- (b) Oxalic acid solutions may be analysed using acid-base titrations. The simplified pH curve for the titration of a solution of oxalic acid against sodium hydroxide is given below.



Four students undertook the titration in different ways:

- Alice used a pH probe to measure the pH after addition of each portion of sodium hydroxide solution
- Brychan used the indicator phenolphthalein in his titration; phenolphthalein has a pH range of 8.2 to 10.0
- Carys used the indicator methyl orange in her titration; methyl orange has a pH range of 3.2 to 4.4
- David used the indicator cresol red in his titration; cresol red has a pH range of 1.8 to 2.8



State and explain which, if any, of the methods chosen will allow the students to find valid data to calculate the concentration of oxalic acid. Briefly explain why oxalic acid has a pH curve of this shape. [6 QER]

*You are not required to carry out any calculations.*

Examiner  
only

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